Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

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***Abstract*—Zakat is a mandatory annual obligation in Islam for wealth redistribution. Blockchain-based zakat platforms have emerged, but existing systems log full donor-recipient transactions publicly, contradicting the Islamic principle of nafs (dignity). This paper presents Zero-Knowledge Zakat (ZKT), a protocol that combines off-chain ZK proofs, nullifier-based double-donation prevention, and selective disclosure. Donors select full anonymity or partial disclosure; both produce SBT receipts. On-chain verification provides institutional transparency while individual transaction data stays private. A Sepolia prototype generates UltraHONK proofs in 247 ms, with end-to-end donation consuming 721,345 gas. Cryptographic privacy and institutional accountability prove compatible, supporting digital zakat that preserves donor and recipient privacy. ZKT contributes a privacy protocol for Islamic social finance that directly addresses nafs protection under maqasid al-shariah.**

***Index Terms*—Aztec Network, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Privacy-Preserving Donations, Sharia Compliance, Zakat, Zero-Knowledge Proofs**

I. Introduction

Applying blockchain to zakat management raises practical difficulties. BAZNAS [1] estimates Indonesia's annual zakat potential at Rp 327 trillion (approximately USD 20 billion), yet actual collection remains far lower.

Research into blockchain zakat systems has advanced. Khan [2] showed blockchain can support decentralized collection and distribution. Nazeri et al. [3] found blockchain features (transparency, traceability, security) align with Malaysian zakat management. Khairi et al. [4] developed a blockchain collection system offering automated disbursement and an immutable audit trail. These systems share a privacy weakness: full donor-recipient transaction records posted on public ledgers expose socioeconomic vulnerabilities. Maqasid al-shariah (objectives of Islamic law) explicitly protects *nafs* (life and dignity) of individuals [5, 6]. Islamic jurisprudence treats donor and recipient privacy in charitable giving as a virtuous act that preserves dignity. Blockchain transparency therefore conflicts with Islamic privacy norms.

Zero-Knowledge Proofs (ZKPs) address this conflict. Hameed et al. [7] and Wen et al. [8] survey ZKP constructions and zk-SNARKs suitable for blockchain. The Aztec Network [9] provides privacy-preserving Ethereum transactions via a hybrid public-private state model with Noir circuits [10] and the UltraHONK prover. No prior work proposes a domain-specific protocol for privacy-preserving zakat. This paper presents Zero-Knowledge Zakat (ZKT), a private transaction model that couples off-chain proving with on-chain verification. The implementation uses Noir circuits for Sharia eligibility checks, nullifiers for double-donation prevention, and a working Sepolia prototype. The work investigates whether Aztec infrastructure serves zakat donations with institutional accountability, whether Noir design patterns encode Sharia eligibility, and whether the protocol resolves nafs-amana tensions under maqasid al-shariah.

This paper contributes a domain-specific privacy protocol for Islamic social finance, featuring a Noir-based circuit implementing nisab and hawl eligibility verification as verifiable predicates, a nullifier mechanism preventing re-claiming without identity exposure, and a tiered donor privacy model. The prototype achieves UltraHONK proof generation averaging 247 ms and end-to-end donation gas costs of 721,345 gas on Ethereum Sepolia.

II. Literature Review

Blockchain-based zakat management has attracted increasing research interest. Khan [2] demonstrated a blockchain-based decentralized zakat collection and distribution platform that eliminates intermediary delays. Khairi et al. [4] developed a zakat collection blockchain system employing ERC-20 tokens to provide an immutable audit trail. Nazeri et al. [3] explored how blockchain features align with zakat management objectives in Malaysia. A recent Emerald study examined the potential and challenges of blockchain technology for zakat institutions [21]. However, these systems share a critical privacy limitation. Complete transparency of donor-recipient relationships on public blockchains enables social stigma, targeted harassment, and pattern analysis attacks [2]. Fauzi et al. [19] confirmed through bibliometric analysis an increasing research trend in Muslim-majority countries. Ascarya [20] demonstrated that Islamic social finance instruments (zakat, infaq, sadaqah, and waqf) play a role in economic recovery, and Unal and Aysan [21] proposed blockchain as a means to harmonize differing Sharia-compliance regimes. Muharam and Osman [27] explored the nexus of blockchain and Islamic social finance for sustainability. Yet none of these works integrate privacy-preserving transaction mechanisms.

On the zero-knowledge proof front, Hameed et al. [7] survey ZKP constructions, and Wen et al. [8] analyze zk-SNARK variants suitable for blockchain deployment. Williams [22] provides a contemporary overview of ZKPs and their expanding role within blockchain technology. Grassi et al. [11] introduce the Poseidon hash function, which reduces constraint count by up to 8× compared to Pedersen Hash on EVM chains. For this work, Noir [10] is selected for its integration with the Aztec Network [9], which provides programmable privacy through a hybrid public-private state model with note hash trees and nullifier mechanisms. Buterin et al. [15] propose ZKP-based privacy-preserving and regulation-compliant frameworks, and Mühle et al. [18] present zk-SNARK-based identity management suitable for KYC integration.

Three research gaps are identified. (1) Existing blockchain-based zakat platforms lack privacy-preserving transaction mechanisms [2, 3]. (2) Private transaction frameworks are designed for general-purpose anonymity without Islamic jurisprudential eligibility criteria [15, 14]. (3) The integration of programmable privacy with Islamic social finance has not been explored [21, 5]. This paper directly addresses all three gaps.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT bridges all three domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8).

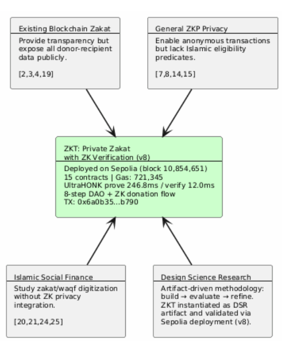

Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

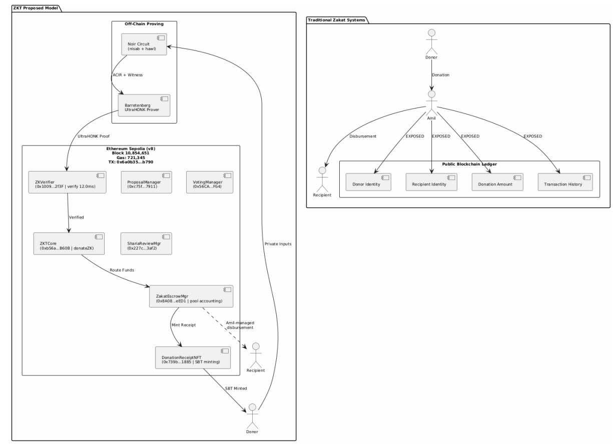

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. Methodology

This study uses a Design Science Research (DSR) approach. Fig. 3 shows the five-phase methodology: Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi/XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with performance of 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis, and Sharia compliance review.

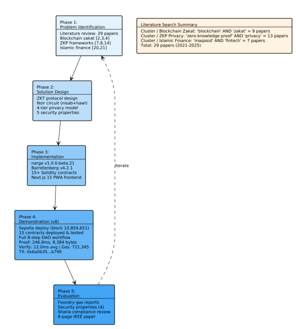

Fig. 3. DSR Methodology with five phases and literature search summary

IV. Results

A. *System Overview and Threat Model*

The ZKT protocol uses a five-actor trust model: donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT PROTOCOL SECURITY PROPERTIES

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain; only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs; recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6].

Fig. 4 shows the three-layer protocol architecture: the off-chain proving pipeline where Noir circuits compile to ACIR and Barretenberg generates UltraHONK proofs at 247 ms, the Sepolia settlement layer hosting ZKTCore and ZakatEscrowManager, and the Next.js PWA frontend with wagmi/XellarKit wallet connectivity.

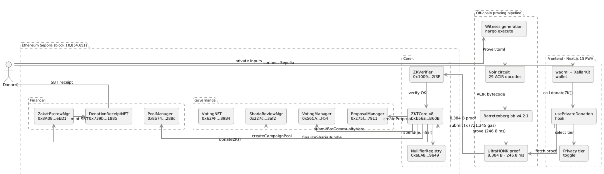

Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

B. System Architecture and Payment Flow

The platform uses a hybrid architecture: Noir circuits on the client side with Ethereum Sepolia for public settlement. Donors pay in stablecoins or ETH. Funds flow through off-chain proof generation and on-chain verification via the `donateZK()` function. Individual donation privacy is maintained through cryptographic proofs, while institutional accountability is achieved through on-chain event emission and nullifier tracking.

A Progressive Web Application handles the donor workflow: wallet connection, privacy tier selection, donation amount entry, proof generation display, and transaction confirmation. Proof generation runs in a Web Worker ensuring that private inputs never leave the donor's device [11].

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end donateZK() gas was measured on the Sepolia testnet at block 10,854,651.

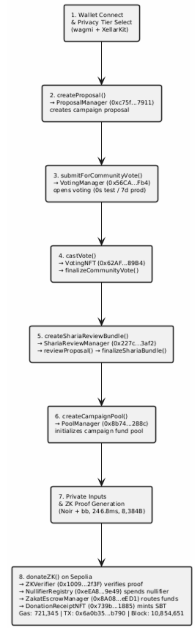

Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

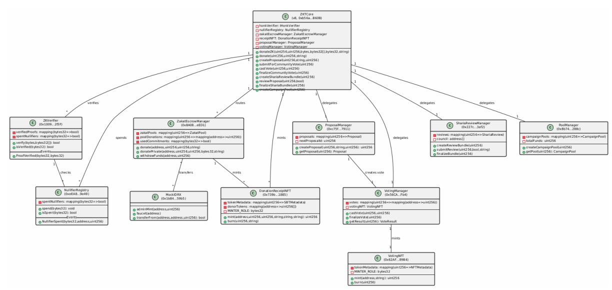

Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCore (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCore	`donate()`	490,187
ZKTCore	`createCampaignPool()`	365,552
ZKTCore	`castVote()`	132,909
ZKTCore	`donateZK()` (e2e)	721,345
ZKTCore	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

E. Security Analysis

Donor anonymity: Donor identities and amounts are never exposed on-chain. Only ZK proofs are publicly verifiable [12], [22]. **Recipient privacy**: Donor-side privacy preserved through ZK proofs and nullifier anonymity. Recipient-side encrypted note disbursement is reserved for future work [16]. **Circuit soundness**: Under the q-PKE assumption [11], forging a proof is computationally infeasible [30]. **Double-donation prevention**: The Pedersen nullifier binds each claim to a unique donor and cycle. The ZKTCore contract ensures atomic rejection of duplicate claims [19].

V. DISCUSSION

A. Sharia Compliance and Deployment

The principle of maqasid al-shariah (objectives of Islamic law) explicitly protects *nafs* (life and dignity) of individuals [5, 6]. In Islamic jurisprudence, preservation of donor and recipient privacy during charitable giving is considered a virtuous act that maintains dignity for both parties [5, 6]. Azizov et al. [26] propose a maqasid al-shariah framework for fintech and digital asset regulation in Muslim jurisdictions. Latang et al. [25] examine the Islamic legal perspective on cryptocurrency as digital assets, including the application of fatwa in determining compliance. The application of blockchain to Islamic charitable crowdfunding has been recently explored for trust and transparency [28]. This research applies privacy-preserving mechanisms to the donor side through ZK proofs and nullifier-based anonymity. Recipient privacy preservation through encrypted private notes and selective disclosure is reserved for future work by the authors. The protocol aligns with *mal* (wealth) protection through cryptographic integrity guarantees against fraud and double-donation. *Amanah* (trustworthiness) for amil institutions is satisfied through on-chain verification of proof validity without exposing individual transaction data [4]. Deployment requires Aztec Network integration, amil staff training, and regulatory alignment with OJK Fintech Sandbox requirements.

B. Limitations and Future Work

Circuit scope is limited to nisab and hawl criteria. Extension to all eight asnaf categories is architecturally accommodated. Currently, zakat disbursement is performed by the amil institution through the ZakatEscrowManager contract without on-chain verifiability of the disbursement to mustahik (recipients). Future work includes issuing private decentralized identifiers (DIDs) to mustahik for on-chain eligibility attestation, making amil disbursements cryptographically verifiable, and implementing a dual governance system combining Sharia Council ZK DAO attestation with Community (Donor) DAO voting. Additional directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [18], [24].

VI. CONCLUSION

This paper presented a Zero-Knowledge Zakat (ZKT) protocol addressing the privacy paradox in blockchain-based zakat systems. An implementation using Noir circuits (29 ACIR opcodes) with UltraHONK proving via Barretenberg v4.2.1 was deployed to Ethereum Sepolia, achieving proof generation in 247 ms and end-to-end donation in 721,345 gas. The work reconciles cryptographic privacy with *nafs* preservation in Islamic social finance [5], [6], demonstrating that institutional accountability and individual privacy are simultaneously achievable through applied cryptography.

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